HPC / HA (slurm) Cluster build

6 nodes setup: utilizing and reconfiguring the 2 prior clusters built for kubernetes poc and the non k8's cluster for the ai llm (gemma3 27b) poc

reconfigured specs for the 2 different distros of debian 13 an suse 16 – it's more complicated due to different distro caveats

debian 3x nodes: 2 cpu's, 8gb ram, 50gb storage
suse 3x nodes: 10 cpu's, 128gb ram, 2 x nodes at 1tb storage, 1 node at 2 tb storage

Test configurations:
Reconfiguration of 2 different intents poc into a HPC/HA cluster proof of concept test build

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// initial prerequisite packages starting with suse, repeat for all the nodes

```
sles16-node1:/home/onemantech # zypper install munge munge-devel slurm slurm-node openmpi5 openmpi5-devel
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 9 recommended packages were automatically selected:
  munge openmpi5-config pmix-devel pmix-plugin-munge rdma-ndd readline-doc slurm-config-man slurm-doc slurm-munge

The following 55 NEW packages are going to be installed:
  hwloc-data hwloc-devel infiniband-diags libefa1 libevent-devel libfabric1 libfreeipmi17 libhns1 libhwloc15 libibmad5 libibnetdisc5 libibumad3 libibverbs
  libibverbs1 libipmimonitoring6 libmana1 libmca_common_dstore1 libmlx4-1 libmlx5-1 libmunge2 libpciaccess-devel libpmix2 libpsm2-2 librdmacm1 libucm0
  libucp0 libucs0 libuct0 libxml2-devel mpi-selector munge munge-devel openmpi5 openmpi5-config openmpi5-devel openmpi5-libs pmix pmix-devel pmix-headers
  pmix-plugin-munge pmix-plugins rdma-core rdma-core-devel rdma-ndd readline-devel readline-doc rsocket slurm slurm-config slurm-config-man slurm-doc
  slurm-munge slurm-node slurm-plugins xz-devel

55 new packages to install.

Package download size:    19.0 MiB

Package install size change:
              |      56.7 MiB  required by packages that will be installed
    56.7 MiB  |  -      0 B    released by packages that will be removed

Backend:  classic_rpmtrans
Continue? [y/n/v/...? shows all options] (y): y
Preloading: pmix-plugin-munge-3.2.5-bp160.1.5.x86_64.rpm [done]
Preloading: libmlx4-1-56.1-160000.2.2.x86_64.rpm [done]
```

// initial prerequisite packages with debian, repeat for all the nodes

```
root@dkm:/home/test# apt-get install munge libmunge-dev slurmd slurm-client libopenmpi-dev
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  autoconf automake autotools-dev binutils binutils-common binutils-x86-64-linux-gnu freeipmi-common gcc gcc-14 gcc-14-x86-64-linux-gnu
  gcc-x86-64-linux-gnu gfortran gfortran-14 gfortran-14-x86-64-linux-gnu gfortran-x86-64-linux-gnu ibverbs-providers libaec0 libamd-comgr2 libamdhip64-5
  libasan8 libb64-0d libbinutils libc-dev-bin libc6-dev libcaf-openmpi-3t64 libcc1-0 libcoarrays-dev libcoarrays-openmpi-dev libcrypt-dev libctf-nobfd0
  libctf0 libevent-core-2.1-7t64 libevent-dev libevent-extra-2.1-7t64 libevent-openssl-2.1-7t64 libevent-pthreads-2.1-7t64 libfabric1 libfreeipmi17
  libgcc-14-dev libgfortran-14-dev libgprofng0 libhdf5-310 libhdf5-hl-310 libhsa-runtime64-1 libhsakmt1 libhwasan0 libhwloc-dev libhwloc-plugins libhwloc15
  libibmad5 libibumad3 libibverbs-dev libibverbs1 libipmimonitoring6 libitm1 libjs-jquery libjs-jquery-ui libjwt2 libllvm17t64 liblsan0 libltdl-dev
  liblua5.1-0 libmariadb3 libmunge2 libnl-3-dev libnl-route-3-dev libnuma-dev liboam-dev liboam1 libopenmpi40 libpmix-dev libpmix2t64 libpsm2-2
  libquadmath0 librdmacm1t64 librocm-smi-dev librocm-smi64-1 libsframe1 libsz2 libtool libtsan2 libubsan1 libucx0 libxnvctrl0 libze1 linux-libc-dev m4
  manpages-dev mariadb-common mysql-common openmpi-bin openmpi-common rpcsvc-proto slurm-wlm-basic-plugins slurm-wlm-elasticsearch-plugin
  slurm-wlm-hdf5-plugin slurm-wlm-influxdb-plugin slurm-wlm-ipmi-plugins slurm-wlm-jwt-plugin slurm-wlm-mysql-plugin slurm-wlm-plugins
  slurm-wlm-rsmi-plugin zlib1g-dev
Suggested packages:
  autoconf-archive gnu-standards autoconf-doc gettext binutils-doc gprofng-gui binutils-gold freeipmi-tools gcc-multilib make flex bison gdb gcc-doc
  gcc-14-multilib gcc-14-doc gcc-14-locales gdb-x86-64-linux-gnu gfortran-multilib gfortran-doc gfortran-14-multilib gfortran-14-doc libc-devtools
  glibc-doc libhwloc-contrib-plugins libjs-jquery-ui-docs libtool-doc openmpi-doc gcj-jdk m4-doc
The following NEW packages will be installed:
  autoconf automake autotools-dev binutils binutils-common binutils-x86-64-linux-gnu freeipmi-common gcc gcc-14 gcc-14-x86-64-linux-gnu
  gcc-x86-64-linux-gnu gfortran gfortran-14 gfortran-14-x86-64-linux-gnu gfortran-x86-64-linux-gnu ibverbs-providers libaec0 libamd-comgr2 libamdhip64-5
  libasan8 libb64-0d libbinutils libc-dev-bin libc6-dev libcaf-openmpi-3t64 libcc1-0 libcoarrays-dev libcoarrays-openmpi-dev libcrypt-dev libctf-nobfd0
  libctf0 libevent-core-2.1-7t64 libevent-dev libevent-extra-2.1-7t64 libevent-openssl-2.1-7t64 libevent-pthreads-2.1-7t64 libfabric1 libfreeipmi17
  libgcc-14-dev libgfortran-14-dev libgprofng0 libhdf5-310 libhdf5-hl-310 libhsa-runtime64-1 libhsakmt1 libhwasan0 libhwloc-dev libhwloc-plugins libhwloc15
  libibmad5 libibumad3 libibverbs-dev libibverbs1 libipmimonitoring6 libitm1 libjs-jquery libjs-jquery-ui libjwt2 libllvm17t64 liblsan0 libltdl-dev
  liblua5.1-0 libmariadb3 libmunge-dev libmunge2 libnl-3-dev libnl-route-3-dev libnuma-dev liboam-dev liboam1 libopenmpi-dev libopenmpi40 libpmix-dev
  libpmix2t64 libpsm2-2 libquadmath0 librdmacm1t64 librocm-smi-dev librocm-smi64-1 libsframe1 libsz2 libtool libtsan2 libubsan1 libucx0 libxnvctrl0 libze1
  linux-libc-dev m4 manpages-dev mariadb-common munge mysql-common openmpi-bin openmpi-common rpcsvc-proto slurm-client slurm-wlm-basic-plugins
  slurm-wlm-elasticsearch-plugin slurm-wlm-hdf5-plugin slurm-wlm-influxdb-plugin slurm-wlm-ipmi-plugins slurm-wlm-jwt-plugin slurm-wlm-mysql-plugin
  slurm-wlm-plugins slurm-wlm-rsmi-plugin slurmd zlib1g-dev
0 upgraded, 108 newly installed, 0 to remove and 0 not upgraded.
Need to get 131 MB of archives.
```

// creating the ssh keys on each node for any to any node ssh communication

```
sles16-node3:/home/onemantech # ssh-keygen -t ed25519 -f /root/.ssh/id_ed25519 -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /root/.ssh/id_ed25519
Your public key has been saved in /root/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:C6bKP25[redacted]os root@sles16-node3
The key's randomart image is:
+--[ED25519 256]--+
|          .++==o |
|         ...+++++|
|    [redacted]   |
|   ..[redacted] .|
| . .+ o      . + |
|  o+o..+.   E .  |
+----[SHA256]-----+
sles16-node3:/home/onemantech # |
```

// creating .ssh config file on the main brain and repeat on the other nodes

```
#manual config file for communication between nodes from the ssh aspect
#
#
host sles16-node1
        hostname 172.[redacted]
        user onemantech
        identityfile ~/.ssh/id_ed25519

host sles16-node2
        hostname 172.[redacted]
        user onemantech
        identityfile ~/.ssh/id_ed25519

host sles16-node3
        hostname 172.[redacted]
        user onemantech
        identityfile ~/.ssh/id_ed25519

host dkm
        hostname 172.[redacted]
        user test
        identityfile ~/.ssh/id_ed25519

host dkw1
        hostname 172.[redacted]
        user test
        identityfile ~/.ssh/id_ed25519

host dkw2
        hostname 172[redacted]
        user test
        identityfile ~/.ssh/id_ed25519
~
```

// edit the “/etc/hosts” file on all the nodes to allow for communication/access via names

```
172.[redacted]     sles16-node1
172.[redacted]     sles16-node2
172.[redacted]     sles16-node3
172.[redacted]     dkm
172.[redacted]     dkw1
172.[redacted]     dkw2
```


// export copy the ssh id on the rest of the nodes (x5 for each node in my case for any to any communication)

```
sles16-node3:/home/onemantech # ssh-copy-id sles16-node2
/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_ed25519.pub"
The authenticity of host '172.[redacted])' can't be established.
ED25519 key fingerprint is SHA256:4nCdS+x[redacted]Aw/m3+7Vai/E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
(onemantech@172.[redacted]) Password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'sles16-node2'"
and check to make sure that only the key(s) you wanted were added.

sles16-node3:/home/onemantech #
```

// verify connection to the other nodes from brain via ssh keys rather than through ip and user/login/pw

```
sles16-node3:/home/onemantech # ssh sles16-node1
Have a lot of fun...
onemantech@sles16-node1:~>
onemantech@sles16-node1:~> exit
logout
Connection to 172.[redacted] closed.
sles16-node3:/home/onemantech #
sles16-node3:/home/onemantech #
sles16-node3:/home/onemantech # ssh sles16-node2
Have a lot of fun...
onemantech@sles16-node2:~>
onemantech@sles16-node2:~>
onemantech@sles16-node2:~> exit
logout
Connection to 172.[redacted] closed.
sles16-node3:/home/onemantech #
```

```
sles16-node3:/home/onemantech # ssh dkm
Linux dkm 6.12.73+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.73-1 (2026-02-17) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Mar  8 10:44:10 2026 from 172.[redacted]
test@dkm:~$
test@dkm:~$ exit
logout
Connection to 172[redacted] closed.
sles16-node3:/home/onemantech #
```

// reverse connection back to brain node

```
test@dkw2:~$ ssh sles16-node3
Have a lot of fun...
onemantech@sles16-node3:~> exit
logout
Connection to 172.[redacted] closed.
test@dkw2:~$
```


// test connection from any and any nodes

```
test@dkw2:~$ ssh dkm
Linux dkm 6.12.73+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.73-1 (2026-02-17) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Mar  8 19:04:35 2026 from 172.[redacted]
test@dkm:~$
```

// configuration verifications for the ssh setup (for me this took the longest time for the any to any node communication in addition to the hours troubleshooting the initial one way communication)

// this file needs to be individualized/customized based on the host

```
#manual config file for communication between nodes from the ssh aspect
#
#
host sles16-node1
        hostname 172.[redacted]
        user onemantech
        identityfile ~/.ssh/sles16-node3

host sles16-node2
        hostname 172.[redacted]
        user onemantech
        identityfile ~/.ssh/sles16-node3

host sles16-node3
        hostname 172.[redacted]
        user onemantech
        identityfile ~/.ssh/sles16-node3

host dkm
        hostname 172.[redacted]
        user test
        identityfile ~/.ssh/sles16-node3
```

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// generally the files that should be seen in the .ssh directory

```
onemantech@sles16-node3:~> ls -lsh ~/.ssh/
total 28K
4.0K -rw-------. 1 onemantech onemantech  587 Mar  8 19:40 authorized_keys
4.0K -rwxr-xr-x. 1 onemantech onemantech  596 Mar  8 18:43 config
8.0K -rw-------. 1 onemantech onemantech 4.1K Mar  8 18:55 known_hosts
4.0K -rw-------. 1 onemantech onemantech 3.4K Mar  8 18:55 known_hosts.old
4.0K -rw-------. 1 onemantech onemantech  419 Mar  8 17:50 sles16-node3
4.0K -rw-r--r--. 1 onemantech onemantech  105 Mar  8 17:50 sles16-node3.pub
onemantech@sles16-node3:~>
```


// I had 6 hosts so 6 keys for the any to any communications

```
test@dkm:~$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAA[redacted] onemantech@sles16-node3
ssh-ed25519 AAAA[redacted] test@dkm
ssh-ed25519 AAAA[redacted] onemantech@sles16-node2
ssh-ed25519 AAAA[redacted] onemantech@sles16-node1
ssh-ed25519 AAAA[redacted] test@dkw1
ssh-ed25519 AAAA[redacted] test@dkw2
test@dkm:~$
```

// edit the chrony.conf on the main brain to act as the source for the rest of the nodes and restart the service

```
sles16-node3:/home/onemantech # vi /etc/chrony.conf
sles16-node3:/home/onemantech #
sles16-node3:/home/onemantech # systemctl restart chronyd.service
```

```
# Allow NTP client access from local network.
allow 172.[redacted]/24

# Serve time even if not synchronized to a time source.
local stratum 10
```

// edit the chrony.conf file on the rest of the nodes to point to the brain as the time source

```
# Manually added this line, to point to the lab server
server 172.[redacted] iburst
```

// suse 16 has the security/firewall that blocks the isolated local time server so need to update the firewall rules

```
sles16-node3:/home/onemantech # firewall-cmd --add-port=123/udp --permanent
success
```

```
sles16-node3:/home/onemantech # firewall-cmd --add-port=323/udp --permanent
success
sles16-node3:/home/onemantech #
sles16-node3:/home/onemantech # firewall-cmd --reload
success
sles16-node3:/home/onemantech #
sles16-node3:/home/onemantech # systemctl restart chronyd.service
sles16-node3:/home/onemantech #
```

```
onemantech@sles16-node3:~> systemctl status chronyd
● chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-03-08 10:40:31 EDT; 9h ago
 Invocation: e9d72fae73d441cbbe54aa88d7fa44ac
       Docs: man:chronyd(8)
             man:chrony.conf(5)
   Main PID: 1149 (chronyd)
      Tasks: 1 (limit: 154533)
        CPU: 616ms
     CGroup: /system.slice/chronyd.service
             └─1149 /usr/sbin/chronyd
```

```
sles16-node1:/home/onemantech # chronyc sources -v

  .-- Source mode  '^' = server, '=' = peer, '#' = local clock.
 / .- Source state '*' = current best, '+' = combined, '-' = not combined,
| /             'x' = may be in error, '~' = too variable, '?' = unusable.
||                                                 .- xxxx [ yyyy ] +/- zzzz
||      Reachability register (octal) -.           |  xxxx = adjusted offset,
||      Log2(Polling interval) --.      |          |  yyyy = measured offset,
||                                \     |          |  zzzz = estimated error.
||                                 |    |           \
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
^* sles16-node3                 10   6    17     3    +63us[  +47us] +/-  355us
sles16-node1:/home/onemantech #
```

```
root@dkw1:/home/test# chronyc sources -v

  .-- Source mode  '^' = server, '=' = peer, '#' = local clock.
 / .- Source state '*' = current best, '+' = combined, '-' = not combined,
| /             'x' = may be in error, '~' = too variable, '?' = unusable.
||                                                 .- xxxx [ yyyy ] +/- zzzz
||      Reachability register (octal) -.           |  xxxx = adjusted offset,
||      Log2(Polling interval) --.      |          |  yyyy = measured offset,
||                                \     |          |  zzzz = estimated error.
||                                 |    |           \
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
^* sles16-node3                 10   6    17     2    +20us[  +65us] +/-  342us
root@dkw1:/home/test#
```

```
sles16-node3:/home/onemantech # chronyc tracking
Reference ID    : 7F7F0101 ()
Stratum         : 10
Ref time (UTC)  : Mon Mar 09 22:49:25 2026
System time     : 0.000000014 seconds fast of NTP time
Last offset     : +0.000000000 seconds
RMS offset      : 0.000000000 seconds
Frequency       : 44.661 ppm slow
Residual freq   : +0.000 ppm
Skew            : 0.000 ppm
Root delay      : 0.000000000 seconds
Root dispersion : 0.000000000 seconds
Update interval : 0.0 seconds
Leap status     : Normal
sles16-node3:/home/onemantech #
```

// copy the munge key from the brain node to the rest of the nodes, in my case it already existed, didn’t have to create it, but it seems like any default install of munge will install it’s own key, so it will need to be erased on the existing hosts

```
sles16-node3:/home/onemantech # ls -lsh /etc/munge/
total 4.0K
4.0K -rw-------. 1 munge munge 1.0K Mar  8 11:23 munge.key
sles16-node3:/home/onemantech #
```

```
sles16-node3:/home/onemantech # scp -v /etc/munge/munge.key dkm:/etc/munge/munge.key
Executing: program /usr/bin/ssh host dkm, user (unspecified), command sftp
debug1: OpenSSH_10.0p2, OpenSSL 3.5.0 8 Apr 2025
debug1: Reading configuration data /usr/etc/ssh/ssh_config
debug1: Reading configuration data /etc/ssh/ssh_config.d/50-suse.conf
debug1: Reading configuration data /etc/crypto-policies/back-ends/openssh.config
debug1: /usr/etc/ssh/ssh_config line 31: include /usr/etc/ssh/ssh_config.d/*.conf matched no files
debug1: /usr/etc/ssh/ssh_config line 33: Applying options for *
debug1: configuration requests final Match pass
debug1: re-parsing configuration
debug1: Reading configuration data /usr/etc/ssh/ssh_config
debug1: Reading configuration data /etc/ssh/ssh_config.d/50-suse.conf
debug1: Reading configuration data /etc/crypto-policies/back-ends/openssh.config
debug1: /usr/etc/ssh/ssh_config line 31: include /usr/etc/ssh/ssh_config.d/*.conf matched no files
debug1: /usr/etc/ssh/ssh_config line 33: Applying options for *
debug1: Connecting to dkm [172[redacted]] port 22.
debug1: Connection established.
debug1: identity file /root/.ssh/id_rsa type -1
debug1: identity file /root/.ssh/id_rsa-cert type -1
```

// created a sample slurm.conf file on the brain and sent/copied it to the other nodes

```
# ====> MANUAL CONFIGURATION OF THIS SLURM.CONF FILE <====
#
# # General Identity
ClusterName=POC-HPC-AI-CLUSTER
SlurmctldHost=sles16-node3

# Security (The Munge you just fixed!)
AuthType=auth/munge
CryptoType=crypto/munge

# File Locations
SlurmctldPidFile=/var/run/slurmctld.pid
SlurmdPidFile=/var/run/slurmd.pid
SlurmdSpoolDir=/var/spool/slurmd
SlurmLogFile=/var/log/slurm/slurmctld.log
SlurmdLogFile=/var/log/slurm/slurmd.log

# Selection and Scheduling
SelectType=select/cons_res
SelectTypeParameters=CR_Core_Memory
SchedulerType=sched/backfill

# Node and Partition Definitions
#
NodeName=dkm CPUs=2 Boards=1 SocketsPerBoard=1 CoresPerSocket=2 ThreadsPerCore=1 RealMemory=7947 State=UNKNOWN
NodeName=dkw1 CPUs=2 Boards=1 SocketsPerBoard=1 CoresPerSocket=2 ThreadsPerCore=1 RealMemory=7947 State=UNKNOWN
NodeName=dkw2 CPUs=2 Boards=1 SocketsPerBoard=1 CoresPerSocket=2 ThreadsPerCore=1 RealMemory=7947 State=UNKNOWN
NodeName=sles16-node1 CPUs=10 Boards=1 SocketsPerBoard=1 CoresPerSocket=10 ThreadsPerCore=1 RealMemory=128812 State=UNKNOWN
NodeName=sles16-node2 CPUs=10 Boards=1 SocketsPerBoard=1 CoresPerSocket=10 ThreadsPerCore=1 RealMemory=128812 State=UNKNOWN
NodeName=sles16-node3 CPUs=10 Boards=1 SocketsPerBoard=1 CoresPerSocket=10 ThreadsPerCore=1 RealMemory=128812 State=UNKNOWN
PartitionName=Main-Cluster Nodes=dkm,dkw1,dkw2,sles16-node1,sles16-node2,sles16-node3 Default=YES MaxTime=INFINITE State=UP
sles16-node3:/home/onemantech #
```

// testing the configuration so far
// attempting to restart the munge service, encountered errors, due to security/permission issues
// ownership of file changed when copied to the rest of the nodes, changed back to owner/group : munge
// repeat for all the nodes

```
root@dkm:/home/test#
root@dkm:/home/test# systemctl restart munge
root@dkm:/home/test#
```

```
root@dkm:/home/test# systemctl restart munge
Job for munge.service failed because the control process exited with error code.
See "systemctl status munge.service" and "journalctl -xeu munge.service" for details.
root@dkm:/home/test#
root@dkm:/home/test# ls -lsh /etc/munge/
total 4.0K
4.0K -rw------- 1 root root 1.0K Mar  9 19:31 munge.key
root@dkm:/home/test#
root@dkm:/home/test# chown munge:munge /etc/munge/munge.key
root@dkm:/home/test#
root@dkm:/home/test# ls -lsh /etc/munge/
total 4.0K
4.0K -rw------- 1 munge munge 1.0K Mar  9 19:31 munge.key
root@dkm:/home/test#
```

// so this is the turning point of everything, after endless hours of playing with the configuration the initial intent of trying to make sles16 the master brain node proved to be fruitless; as with any iterative testing, test a known good system or a new system, and all efforts were more fruitful with same efforts on debian

// meaning a debian node is now configured as the master node

// everything from source package compilation to firewall and various configuration file setups, and all the linux voodoo dark arts...sles16 did not want to play nice with being a master node, so it was left demoted as a worker node :-) revised the slurm.conf file on debian, added a second node as the backup controller

```
root@dkm:/tmp# cat /etc/slurm/slurm.conf
# # General Identity
ClusterName=POC-HPC-AI-CLUSTER
SlurmctldHost=dkm
SlurmctldHost=dkw2

# Security (The Munge you just fixed!)
AuthType=auth/munge
CryptoType=crypto/munge

# File Locations
SlurmctldPidFile=/var/run/slurmctld.pid
SlurmdPidFile=/var/run/slurmd.pid
SlurmdSpoolDir=/var/lib/slurmd
StateSaveLocation=/mnt/cluster-share/slurm/slurmctld
SlurmctldLogFile=/var/lib/slurmctld.log
SlurmdLogFile=/var/lib/slurmd.log
```

// pushed out the configurations to the other nodes restarted the munge / slurmd / slurmctld services

// the cluster is up and recognizing all the nodes

```
root@dkm:/tmp# sinfo -Nel
Wed Mar 11 20:58:29 2026
NODELIST      NODES     PARTITION       STATE CPUS    S:C:T MEMORY TMP_DISK WEIGHT AVAIL_FE REASON
dkm               1 Main-Cluster*        idle 2       1:2:1   7947        0      1   (null) none
dkw1              1 Main-Cluster*        idle 2       1:2:1   7947        0      1   (null) none
dkw2              1 Main-Cluster*        idle 2       1:2:1   7947        0      1   (null) none
sles16-node1      1 Main-Cluster*        idle 10     1:10:1 128812        0      1   (null) none
sles16-node2      1 Main-Cluster*        idle 10     1:10:1 128812        0      1   (null) none
sles16-node3      1 Main-Cluster*        idle 10     1:10:1 128812        0      1   (null) none
root@dkm:/tmp# 
```

```
root@dkm:/tmp# srun -N6 uptime
 21:00:19 up 1 day,  5:45,  1 user,  load average: 0.10, 0.08, 0.05
 21:00:19 up  8:22,  1 user,  load average: 0.68, 0.71, 0.64
 21:00:19 up 1 day, 17:46,  1 user,  load average: 0.07, 0.06, 0.07
 21:00:19  up   8:29,  3 users,  load average: 0.02, 0.01, 0.03
 21:00:19  up 1 day 18:55,  3 users,  load average: 0.03, 0.09, 0.09
 21:00:19  up 1 day 16:37,  3 users,  load average: 0.01, 0.00, 0.00
root@dkm:/tmp# 
```

// here’s where we can see the HPC/HA aspect of this proof of concept clusters

// clustered node execution was already demonstrated in the previous picture, but here going forward will be the job running and then rebooting the primary controller node and verifying the second controller taking over and resuming the job, in addition to the high availability (HA) resiliency of the task still being alive even with a node going down

```
root@dkm:/tmp# srun -N6 -t 00:10:00 -v -k bash -c "sleep 300; uptime" &
[1] 281278
root@dkm:/tmp# srun: defined options
srun: -------------------- --------------------
srun: no-kill             : set
srun: nodes               : 6
srun: time                : 00:10:00
srun: verbose             : 1
srun: -------------------- --------------------
srun: end of defined options
srun: Nodes dkm,dkw[1-2],sles16-node[1-3] are ready for job
srun: jobid 29: nodes(6):`dkm,dkw[1-2],sles16-node[1-3]', cpu counts: 1(x6)
srun: CpuBindType=(null type)
srun: launching StepId=29.0 on host dkm, 1 tasks: 0
srun: launching StepId=29.0 on host dkw1, 1 tasks: 1
srun: launching StepId=29.0 on host dkw2, 1 tasks: 2
srun: launching StepId=29.0 on host sles16-node1, 1 tasks: 3
srun: launching StepId=29.0 on host sles16-node2, 1 tasks: 4
srun: launching StepId=29.0 on host sles16-node3, 1 tasks: 5
srun: topology/default: init: topology Default plugin loaded
srun: Node dkm, 1 tasks started
srun: Node dkw1, 1 tasks started
srun: Node dkw2, 1 tasks started
srun: Node sles16-node1, 1 tasks started
srun: Node sles16-node3, 1 tasks started
srun: Node sles16-node2, 1 tasks started
```

// here we can see all the nodes are active (not idle) for the state, we can see the job (29) working

```
root@dkm:/tmp#
root@dkm:/tmp# sinfo -Nel
Wed Mar 11 21:21:55 2026
NODELIST      NODES     PARTITION       STATE CPUS    S:C:T MEMORY TMP_DISK WEIGHT AVAIL_FE REASON
dkm               1 Main-Cluster*       mixed 2       1:2:1   7947        0      1   (null) none
dkw1              1 Main-Cluster*       mixed 2       1:2:1   7947        0      1   (null) none
dkw2              1 Main-Cluster*       mixed 2       1:2:1   7947        0      1   (null) none
sles16-node1      1 Main-Cluster*       mixed 10     1:10:1 128812        0      1   (null) none
sles16-node2      1 Main-Cluster*       mixed 10     1:10:1 128812        0      1   (null) none
sles16-node3      1 Main-Cluster*       mixed 10     1:10:1 128812        0      1   (null) none
root@dkm:/tmp#
root@dkm:/tmp#
root@dkm:/tmp# squeue
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
                29 Main-Clus     bash     root  R       3:37      6 dkm,dkw[1-2],sles16-node[1-3]
root@dkm:/tmp#
root@dkm:/tmp# |
```

// here’s the manual reboot of the primary controller

```
root@dkm:/tmp# reboot -f
Rebooting.
Read from remote host 172.[redacted] Connection reset by peer
Connection to 172.[redacted] closed.
client_loop: send disconnect: Broken pipe
onemantech@hp-17-cn3582nr:~>
onemantech@hp-17-cn3582nr:~> ssh test@172.[redacted]
test@172.[redacted] password:
Linux dkm 6.12.73+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.73-1 (2026-02-17) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Mar 11 12:38:08 2026 from 172.[redacted]
test@dkm:~$ su
Password:
root@dkm:/home/test#
root@dkm:/home/test#
```

// this is the secondary controller taking over

```
[2026-03-11T21:24:24.001] creating clustername file: ClusterName=poc-hpc-ai-cluster ClusterID=3812
[2026-03-11T21:24:50.418] No memory enforcing mechanism configured.
[2026-03-11T21:24:50.505] Recovered state of 6 nodes
[2026-03-11T21:24:50.512] Recovered JobId=29 StepId=0
[2026-03-11T21:24:50.512] Recovered JobId=29 Assoc=0
[2026-03-11T21:24:50.512] Recovered information about 1 jobs
[2026-03-11T21:24:50.515] Recovered state of 1 partitions
[2026-03-11T21:24:50.516] select/cons_tres: part_data_create_array: select/cons_tres: preparing for 1 partitions
[2026-03-11T21:24:50.519] Recovered state of 0 reservations
[2026-03-11T21:24:50.521] select/cons_tres: select_p_reconfigure: select/cons_tres: reconfigure
[2026-03-11T21:24:50.521] select/cons_tres: part_data_create_array: select/cons_tres: preparing for 1 partitions
[2026-03-11T21:24:50.522] Running as primary controller
```

```
[2026-03-11T21:24:50.551] validate_node_specs: Node dkm unexpectedly rebooted boot_time=1773278678 last response=1773278548
[2026-03-11T21:24:50.551] error: Removing failed node dkm from JobId=29
[2026-03-11T21:24:50.556] Killing JobId=29 StepId=0 on failed node dkm
```

```
root@dkw2:/tmp# sinfo -Nel
Wed Mar 11 21:23:30 2026
slurm_load_partitions: Slurm backup controller in standby mode
root@dkw2:/tmp#
root@dkw2:/tmp# sinfo -Nel
Wed Mar 11 21:24:29 2026
slurm_load_partitions: Socket timed out on send/recv operation
root@dkw2:/tmp# sinfo -Nel
Wed Mar 11 21:24:48 2026
NODELIST      NODES   PARTITION      STATE CPUS    S:C:T MEMORY TMP_DISK WEIGHT AVAIL_FE REASON
dkm               1 Main-Cluster*     down 2       1:2:1   7947        0      1   (null) Node unexpectedly re
dkw1              1 Main-Cluster*    mixed 2       1:2:1   7947        0      1   (null) none
dkw2              1 Main-Cluster*    mixed 2       1:2:1   7947        0      1   (null) none
sles16-node1      1 Main-Cluster*    mixed 10     1:10:1 128812        0      1   (null) none
sles16-node2      1 Main-Cluster*    mixed 10     1:10:1 128812        0      1   (null) none
sles16-node3      1 Main-Cluster*    mixed 10     1:10:1 128812        0      1   (null) none
```

// once the primary node for the slurmctld was back up and service restarted it resume being the primary controller again and the secondary controller resumed being on standby in the background mode

```
[2026-03-11T21:31:04.527] slurmctld version 24.11.5 started on cluster poc-hpc-ai-cluster(3812)
[2026-03-11T21:31:33.134] _read_slurm_cgroup_conf: No cgroup.conf file (/etc/slurm/cgroup.conf), using defaults
[2026-03-11T21:31:33.134] No memory enforcing mechanism configured.
[2026-03-11T21:31:33.269] Recovered state of 6 nodes
[2026-03-11T21:31:33.269] Down nodes: dkm
[2026-03-11T21:31:33.382] Recovered JobId=29 Assoc=0
[2026-03-11T21:31:33.384] Recovered information about 1 jobs
[2026-03-11T21:31:33.384] select/cons_tres: part_data_create_array: select/cons_tres: preparing for 1 partitions
[2026-03-11T21:31:33.521] Recovered state of 0 reservations
[2026-03-11T21:31:33.630] select/cons_tres: select_p_reconfigure: select/cons_tres: reconfigure
[2026-03-11T21:31:33.630] select/cons_tres: part_data_create_array: select/cons_tres: preparing for 1 partitions
[2026-03-11T21:31:33.632] Running as primary controller
```

```
[2026-03-11T21:25:37.217] step_partial_comp: JobId=29 StepID=0 invalid; this step may have already completed
[2026-03-11T21:25:37.221] step_partial_comp: JobId=29 StepID=0 invalid; this step may have already completed
[2026-03-11T21:25:37.277] step_partial_comp: JobId=29 StepID=0 invalid; this step may have already completed
[2026-03-11T21:25:37.317] step_partial_comp: JobId=29 StepID=0 invalid; this step may have already completed
[2026-03-11T21:25:37.344] step_partial_comp: JobId=29 StepID=0 invalid; this step may have already completed
[2026-03-11T21:25:50.004] SchedulerParameters=requeue_setup_env_fail
[2026-03-11T21:29:21.000] Time limit exhausted for JobId=29
[2026-03-11T21:31:04.552] Performing RPC: REQUEST_CONTROL
[2026-03-11T21:31:04.554] Terminate signal SIGTERM received
[2026-03-11T21:31:33.000] error: _slurm_rpc_request_control: REQUEST_CONTROL reply but backup not completely done relinquishing control.
[2026-03-11T21:31:33.002] Saving all slurm state
[2026-03-11T21:31:33.227] slurmctld running in background mode
```

// after the duration of this test case, the nodes return to being idle and ready for the next tasks

// the primary node is back up, but listed as down and not ready for tasks

// brought the primary node back up as a worker node to be able to participate in the cluster activities

```
root@dkm:/home/test# sinfo -Nel
Wed Mar 11 21:44:48 2026
NODELIST       NODES      PARTITION       STATE CPUS    S:C:T MEMORY TMP_DISK WEIGHT AVAIL_FE REASON
dkm                1 Main-Cluster*        down 2        1:2:1   7947        0      1   (null) Node unexpectedly re
dkw1               1 Main-Cluster*        idle 2        1:2:1   7947        0      1   (null) none
dkw2               1 Main-Cluster*        idle 2        1:2:1   7947        0      1   (null) none
sles16-node1       1 Main-Cluster*        idle 10      1:10:1 128812        0      1   (null) none
sles16-node2       1 Main-Cluster*        idle 10      1:10:1 128812        0      1   (null) none
sles16-node3       1 Main-Cluster*        idle 10      1:10:1 128812        0      1   (null) none
root@dkm:/home/test#
root@dkm:/home/test#
root@dkm:/home/test# scontrol update nodename=dkm state=resume
root@dkm:/home/test#
root@dkm:/home/test# sinfo -Nel
Wed Mar 11 21:45:49 2026
NODELIST       NODES      PARTITION       STATE CPUS    S:C:T MEMORY TMP_DISK WEIGHT AVAIL_FE REASON
dkm                1 Main-Cluster*        idle 2        1:2:1   7947        0      1   (null) Node unexpectedly re
dkw1               1 Main-Cluster*        idle 2        1:2:1   7947        0      1   (null) none
dkw2               1 Main-Cluster*        idle 2        1:2:1   7947        0      1   (null) none
sles16-node1       1 Main-Cluster*        idle 10      1:10:1 128812        0      1   (null) none
sles16-node2       1 Main-Cluster*        idle 10      1:10:1 128812        0      1   (null) none
sles16-node3       1 Main-Cluster*        idle 10      1:10:1 128812        0      1   (null) none
root@dkm:/home/test# |
```

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So this Cluster build proof of concept for HPC/HA of multiple nodes, the additional challenge of it being a heterogeneous environment with two flavors of linux being sles16 and debian 13, was successful with some detours. Maybe sles16 was too new and too hardened that it was out of scope for the time/effort to spend on debugging it for master node configuration, or somewhere down the line this particular node was too poisoned/corrupted with configurations.

Demonstrated 2 brain, nodes serving as master/secondary control of the slurm daemons, with failover (HA) of one brain going down and the job was still resilient (HA) still processing and not dying, while the primary brain recovered and failback once the node was back up and secondary node returned to standby mode

// Here's to all the unsung heroes (efforts, sleepless nights, hours of debugging and performing linux dark arts)

// matching the slurm package internal revisions from different distro packaging

// matching/changing all the system id/owners of permissions/folders

// setting the chronyc/ntp time sources for synching across all the nodes

// testing all the parameters of the slurm.conf files to just have a functional instance to start without errors

// the testing of all the backend ssh key configurations for talking to each other without manual logins

// the manual build from source of packages

// preparing additional build packages to rebuild the slurm package so all the versions match

```
sles16-node1:/home/onemantech # slurmd -V
slurm 24.11.3
sles16-node1:/home/onemantech #
```

```
root@dkw1:/home/test# slurmd -V
slurm-wlm 24.11.5
```

```
root@dkw1:/home/test# apt-get install build-essential git pkg-config libmunge-dev libmunge2 libssl-dev
-dev libcgroup-dev
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libmunge-dev is already the newest version (0.5.16-1).
libmunge2 is already the newest version (0.5.16-1).
libmunge2 set to manually installed.
python3 is already the newest version (3.13.5-1).
python3 set to manually installed.
zlib1g-dev is already the newest version (1:1.3.dfsg+really1.3.1-1+b1).
zlib1g-dev set to manually installed.
libhwloc-dev is already the newest version (2.12.0-4).
libhwloc-dev set to manually installed.
The following additional packages will be installed:
  dpkg-dev fakeroot g++ g++-14 g++-14-x86-64-linux-gnu g++-x86-64-linux-gnu git-man libalgorithm-diff-p
  libalgorithm-merge-perl libcap-dev libcgroup3 libdpkg-perl liberror-perl libfakeroot libfile-fcntlloc
  libsystemd-dev make patch pkgconf pkgconf-bin
Suggested packages:
  debian-keyring debian-tag2upload-keyring g++-multilib g++-14-multilib gcc-14-doc git-doc git-email gi
  libssl-doc libstdc++-14-doc make-doc ed diffutils-doc
The following NEW packages will be installed:
  build-essential dpkg-dev fakeroot g++ g++-14 g++-14-x86-64-linux-gnu g++-x86-64-linux-gnu git git-man
  libalgorithm-merge-perl libcap-dev libcgroup-dev libcgroup3 libdbus-1-dev libdpkg-perl liberror-perl
```

// the manual package compilation/build from source

```
root@dkw1:/home/test/Downloads/slurm-slurm-24-11-5-1# ./configure --prefix=/usr --with-munge
configure: WARNING: unrecognized options: --with-shared-lib
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking target system type... x86_64-pc-linux-gnu
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for a race-free mkdir -p... /usr/bin/mkdir -p
checking for gawk... no
checking for mawk... mawk
checking whether make sets $(MAKE)... yes
checking whether make supports nested variables... yes
checking whether to enable maintainer-specific portions of Makefiles... no
checking whether to include rpath in build... yes
checking whether make supports the include directive... yes (GNU style)
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether gcc accepts -g... yes
checking for gcc option to enable C11 features... none needed
checking whether gcc understands -c and -o together... yes
checking dependency style of gcc... gcc3
checking for mysql_config... no
checking for mariadb_config... no
configure: WARNING: *** mysql_config not found. Evidently no MySQL development libs installed
checking for gcc... (cached) gcc
checking whether the compiler supports GNU C... (cached) yes
```

```
root@dkw1:/home/test/Downloads/slurm-slurm-24-11-5-1# make -j$(nproc)
make  all-recursive
make[1]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1'
Making all in auxdir
make[2]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/auxdir'
make[2]: Nothing to be done for 'all'.
make[2]: Leaving directory '/home/test/Downloads/slurm-slurm-24-11-5-1/auxdir'
Making all in src
make[2]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src'
Making all in api
make[3]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/api'
(echo "{ global:";    \
 echo "    islurm_*;"; \
 echo "    sack_*;"; \
 echo "    slurm_*;"; \
 echo "    slurmdb_*;"; \
 echo "    working_cluster_rec;"; \
 echo "    node_record_count;"; \
 echo "    active_node_record_count;"; \
 echo "    node_record_table_ptr;"; \
 echo "   local: *;"; \
 echo "};") > version.map
(echo "{ global: *; };") > full_version.map
make[4]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/api'
/bin/bash ../../libtool  --tag=CC   --mode=compile gcc -DHAVE_CONFIG_H -I. -I../.. -I../../sl
ame-pointer -pthread -ggdb3 -Wall -g -O1 -fno-strict-aliasing -MT allocate.lo -MD -MP -MF .de
/bin/bash ../../libtool  --tag=CC   --mode=compile gcc -DHAVE_CONFIG_H -I. -I../.. -I../../sl
ame-pointer -pthread -ggdb3 -Wall -g -O1 -fno-strict-aliasing -MT allocate_msg.lo -MD -MP -MF
libtool: compile:  gcc -DHAVE_CONFIG_H -I. -I../.. -I../../slurm -I../.. -DNUMA_VERSION1_COMP
```

// finishing the compiling along with addressing all dependencies along the way ...

```
root@dkw1:/home/test/Downloads/slurm-slurm-24-11-5-1# make install
Making install in auxdir
make[1]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/auxdir'
make[2]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/auxdir'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/home/test/Downloads/slurm-slurm-24-11-5-1/auxdir'
make[1]: Leaving directory '/home/test/Downloads/slurm-slurm-24-11-5-1/auxdir'
Making install in src
make[1]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src'
Making install in api
make[2]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/api'
make[3]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/api'
make[4]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/common'
make[4]: 'libcommon.la' is up to date.
make[4]: Leaving directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/common'
make[4]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/interfaces'
make[4]: 'libcommon_interfaces.la' is up to date.
make[4]: Leaving directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/interfaces'
make[4]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/conmgr'
make[4]: 'libconmgr.la' is up to date.
make[4]: Leaving directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/conmgr'
make[3]: Leaving directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/api'
make  install-am
make[3]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/api'
make[4]: Entering directory '/home/test/Downloads/slurm-slurm-24-11-5-1/src/common'
make[4]: 'libcommon.o' is up to date.
```

This is complete for the scope of the test case scenario, there might be an update for job task failover to another node or resuming on another node or if configurations of getting the sles16 node to be a functional master node, but for all intents and purposes, the scope of work and completion is satisfactory for me :-)

The electricity bill this month…   :`(

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